



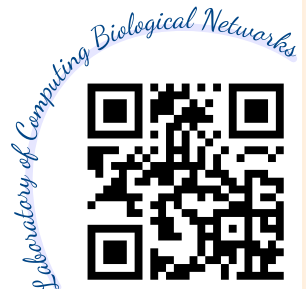
Lecture 13

2025-11-26

Entropy of spike train; Networks of neurons

- Online slides: lec13-networks.html
- Code: [code13.ipynb](#)
- Homework: [hw13.pdf](#), [hw13-data.npz](#)

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Entropy for responses beyond firing rates

– Spike-time characterization of neuronal responses

- Usually much more complex but necessary, because firing rates are incomplete and can seriously underestimate information.
- Careful analysis requires significant amount of statistical power → needs long-duration recordings with many action potentials.
- Measurements typically grow linearly with duration → characterized by entropy/information rates (per unit time or per spike).

Independent inter-spike intervals

– as in renewable processes

With distribution $p[\tau]$ and resolution $\Delta\tau$, entropy per interval:

$$-p[\tau] \log_2 (p[\tau] \Delta\tau)$$

Expected number of intervals for duration T with firing rate $\langle r \rangle$ is $\langle r \rangle T$. It provides an upper bound for entropy rate:

$$\dot{H} \leq -\langle r \rangle \int_0^\infty d\tau p[\tau] \log_2 (p[\tau] \Delta\tau) \quad (4.52)$$

Example: homogeneous Poisson process

For distribution $p[\tau] = \langle r \rangle \exp(-\langle r \rangle \tau)$, we get

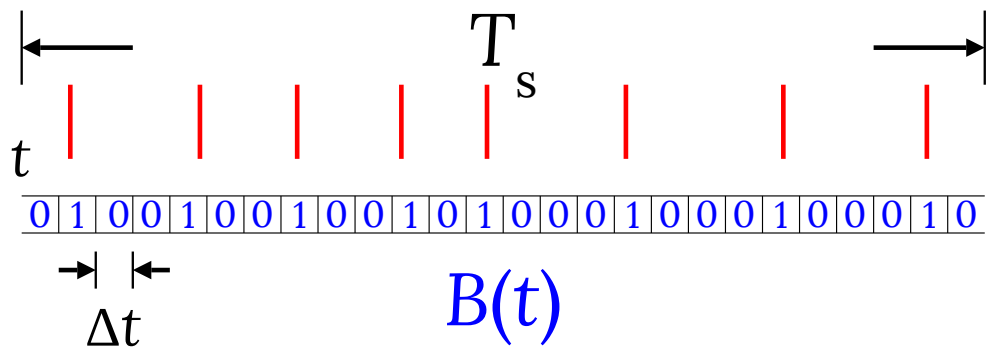
$$\dot{H} = \frac{\langle r \rangle}{\ln 2} (1 - \ln(\langle r \rangle \Delta\tau))$$

→ upper bound for given $p[r]$.

Correlated inter-spike interval

– need to consider subunit with multiple intervals

Fixed duration T_s , spike sequences scheme, Strong et al (1998)



Binarize with small bin size Δt into bit sequence B . Spike-train entropy rate:

$$\dot{H} = -\frac{1}{T_s} \sum_B P[B] \log_2 P[B]$$

Correction for finite T_s decrease with $1/T_s$ for large T_s . Noise entropy is calculated from $P[B(t)]$ obtained with repeated trials under same stimuli at fixed time. Full noise rate:

$$\dot{H} = -\frac{\Delta t}{T} \sum_t \left(\frac{1}{T_s} \sum_B P[B(t)] \log_2 P[B(t)] \right) \quad (4.55)$$

Example: H1 visual neuron in fly

Stimulus: randomly moving visual image

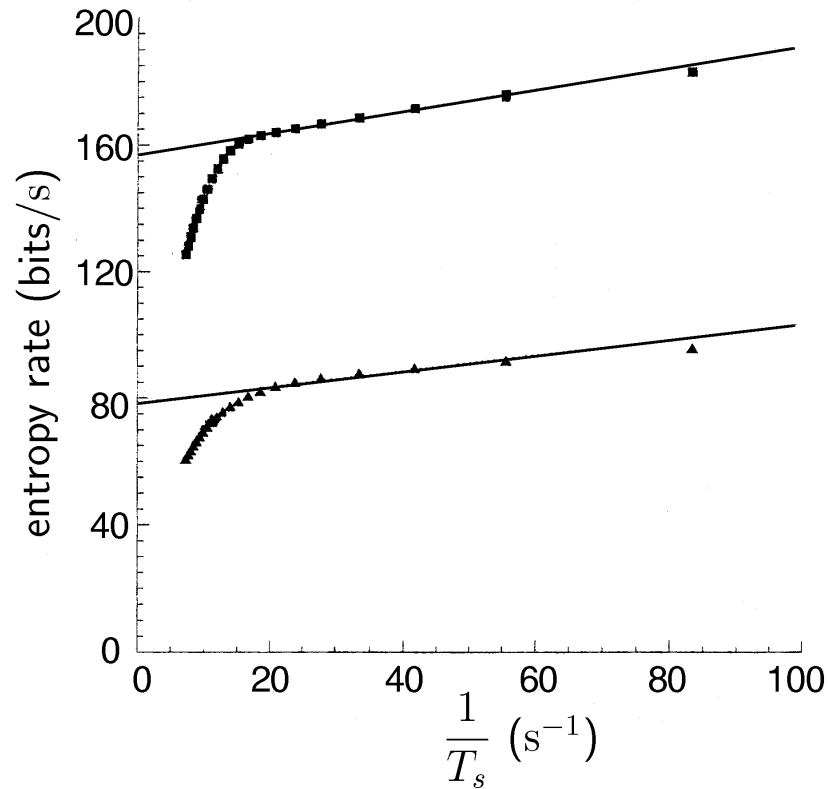


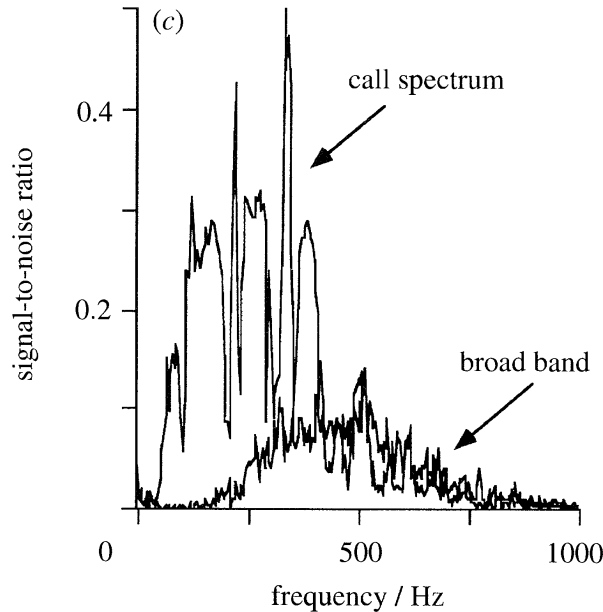
Figure 4.7 from D&A. Upper trace of filled circle: full spike-train entropy rate. Lower trace, spike-train noise entropy rate. Resolution $\Delta t = 3$ ms.

Information rate = $157 - 79 = 78$ bits/s or 1.8 bits/spike.

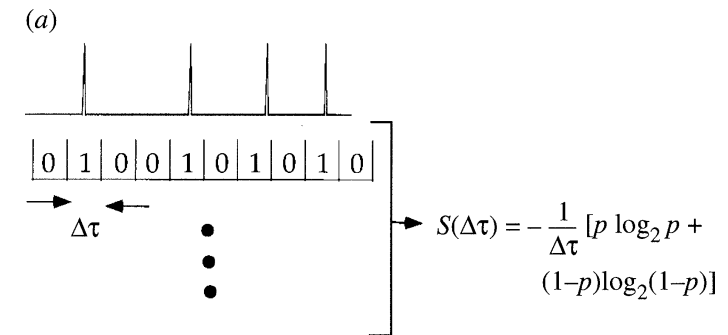
The leading terms on binning resolution $\log_2 \Delta t$ from both contributions cancel. Nondivergent dependency on Δt is expected to increase with decrease of Δt until Δt matches the natural degree of noise jitter in the spike timing.

Example: Peripheral auditory neuron of bullfrog

— Rieke et al. (1995)

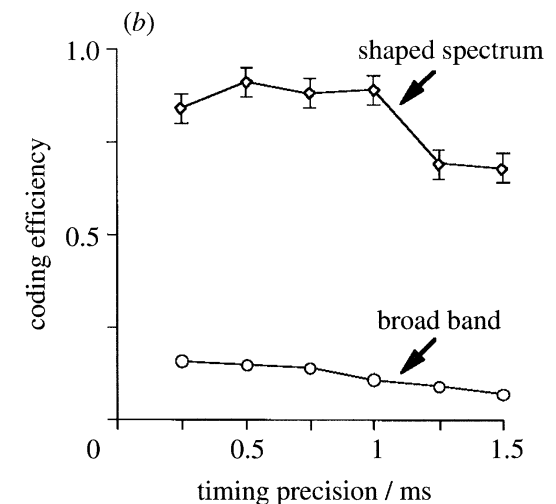


Compare the information conveyed by spike trains in response to different stimuli to reveal stimulus-specific aspects of neural encoding.



Single neurons in a peripheral auditory organ

- broadband noise $\Rightarrow$ 46 bits per second (1.4 bits per spike)
- noise with spectrum of frog calls $\Rightarrow$ 133 bits per second (7.8 bits per spike)



Applications of network models

Allow exploration of the computational potential of synaptic connectivity, through analysis and simulations

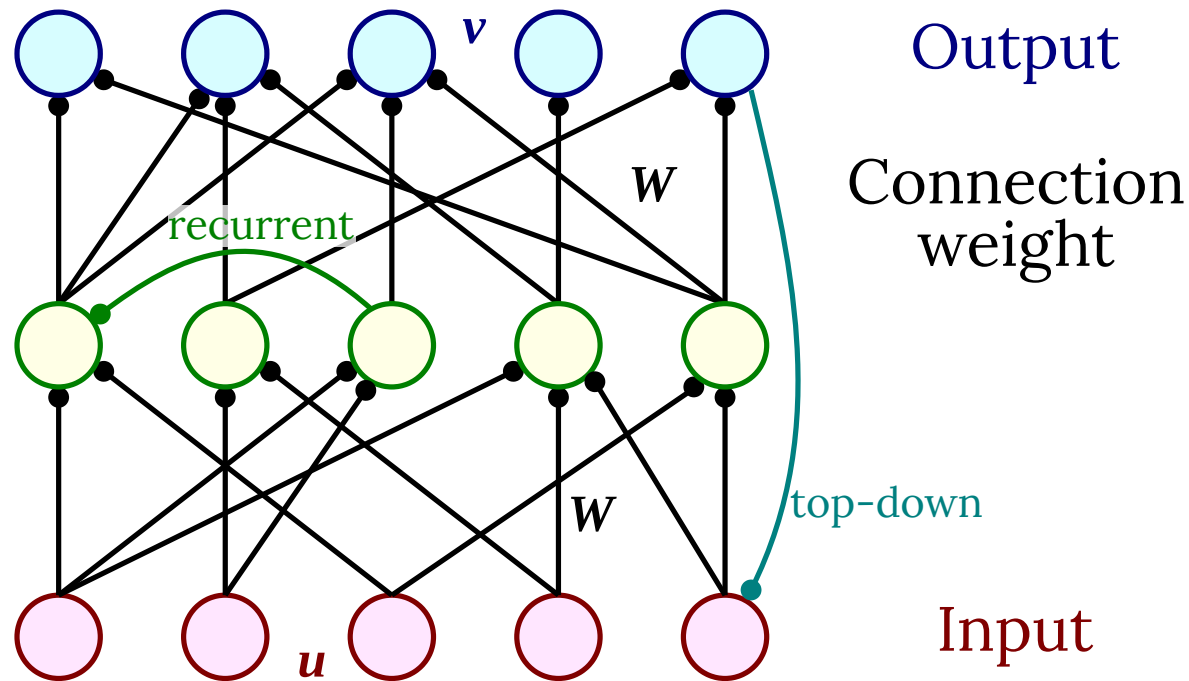
Example tasks

- Coordinate transformation in visually guided reaching
- Selective amplification leading to models of simple & complex cells in VCI
- Integration as a model of short-term memory
- Noise reduction
- Input selection
- Gain modulation
- Associative memory

Network of neurons

Connectivity of neurons:

- Feedforward connection
- Recurrent connection
- Top-down connection



Number of synapses: Feedforward $\approx$ Top-down $<$ Recurrent

Synaptic current from presynaptic activity

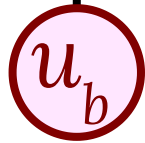
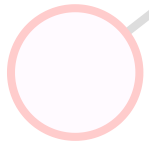
$$w_b \sum_{t_i < t} K_s(t - t_i) = w_b \int_{-\infty}^t d\tau K_s(t - \tau) \rho_b(\tau)$$

postsynaptic
neuron

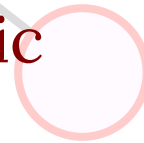


w_b : Synaptic weight with presynaptic neuron b
 ρ_b : spike train of b

$$\rho_b(t) = \sum_i \delta(t_i - t)$$



presynaptic
neuron



w_b

$K_s(t)$: Synaptic kernel

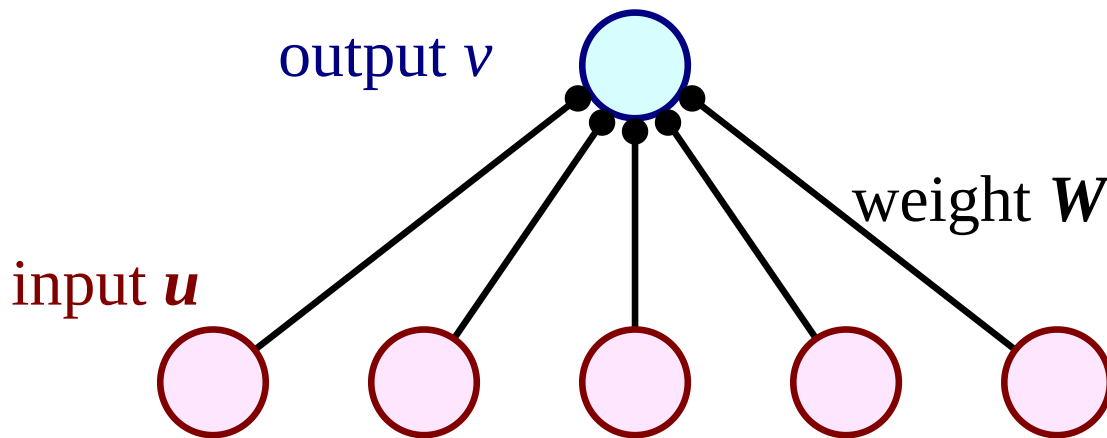
Total current

$$I_s = \sum_{b=1}^{N_u} w_b \int_{-\infty}^t d\tau K_s(t - \tau) \rho_b(\tau)$$

All synapses to firing-rate modeling

$$I_s = \sum_{b=1}^{N_u} w_b \int_{-\infty}^t d\tau K_s(t - \tau) \rho_b(\tau)$$
$$\approx \sum_{b=1}^{N_u} w_b \int_{-\infty}^t d\tau K_s(t - \tau) u_b(\tau)$$

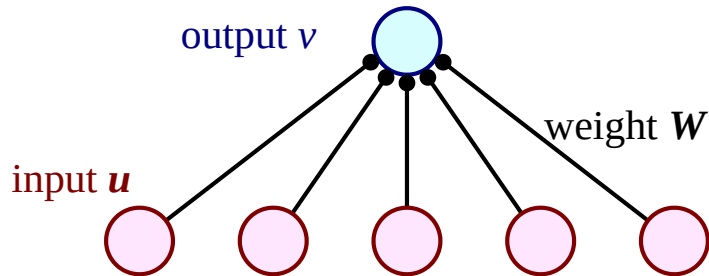
Becomes exact after trial average



$\rho_b(t)$: spike train

$u_b(t)$: firing rate

Exponential kernel for presynaptic action



$$K_s(t) = \frac{1}{\tau_s} \exp \frac{-t}{\tau_s}$$

Apply $\frac{d}{dt}$ to $\rightarrow I_s = \sum_{b=1}^{N_u} w_b \int_{-\infty}^t d\tau K_s(t - \tau) u_b(\tau)$

$$\tau_s \frac{dI_s}{dt} = -I_s + \sum_{b=1}^{N_u} w_b u_b = -I_s + \mathbf{w} \cdot \mathbf{u}$$

For fast synapse $\tau_s \rightarrow 0$ (compared to the dynamics of $\mathbf{u}$), we have $I_s = \mathbf{w} \cdot \mathbf{u}$

Postsynaptic firing rate

Asymptotic firing for slow I_s :

$$v_{\infty} = F(I_s)$$

Low-pass filtering I_s for a time-dependent v

$$\tau_r \frac{dv}{dt} = -v + F(I_s(t))$$

Fast v compared to I

$$\begin{aligned} v &= F(I_s) \\ &\simeq F(\mathbf{w} \cdot \mathbf{u}) \end{aligned}$$

Time scales of two dynamics

Fast postsynaptic rate dynamics

$$\tau_s \frac{dI_s}{dt} = -I_s + \mathbf{w} \cdot \mathbf{u} \quad \text{with} \quad v = F(I_s)$$

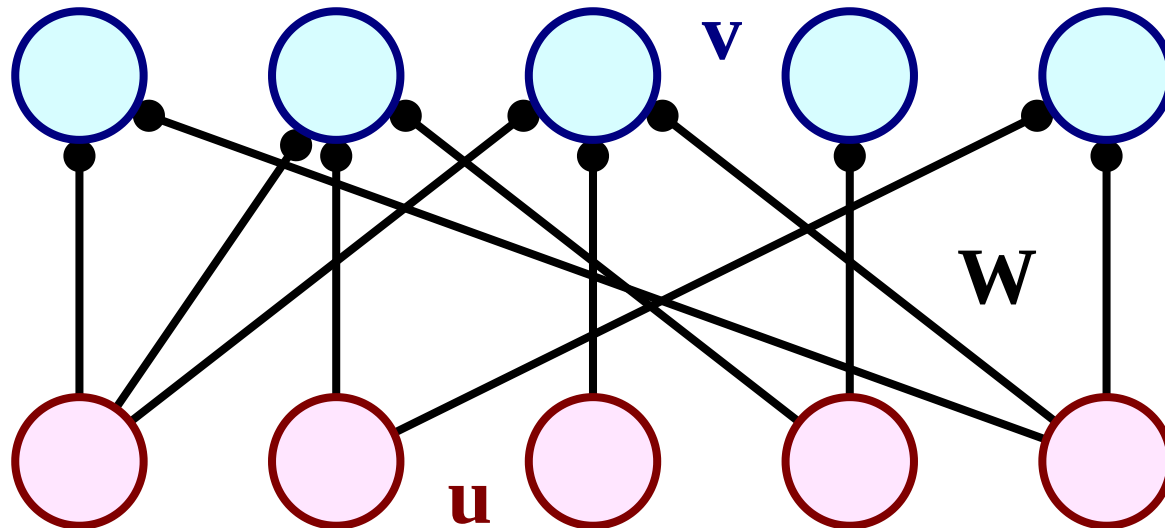
Fast synaptic current dynamics

$$\tau_r \frac{dv}{dt} = -v + F(\mathbf{w} \cdot \mathbf{u})$$

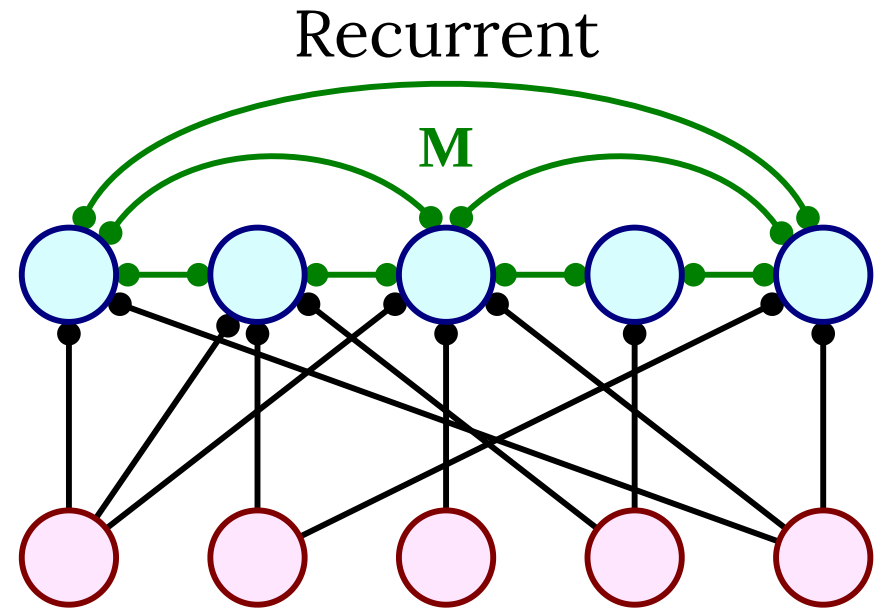
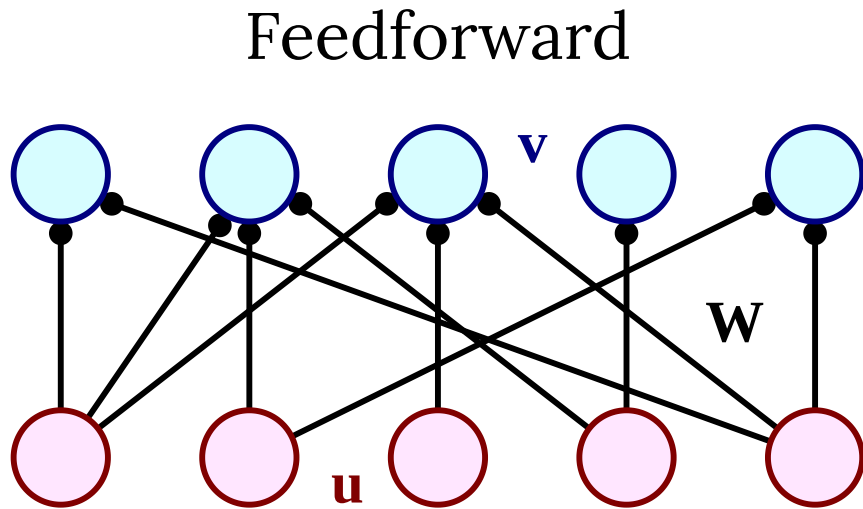
Multiple output neurons

$$\tau_r \frac{d\mathbf{v}}{dt} = -\mathbf{v} + \mathbf{F}(\mathbf{W} \cdot \mathbf{u})$$

Output $\mathbf{v}$ and function $\mathbf{F}$ become vectors while weight $\mathbf{W}$ becomes a rank-two tensor.



Feedforward and recurrent networks



$$\begin{aligned}\tau_r \frac{dv}{dt} &= -\mathbf{v} + \mathbf{F}(\mathbf{W} \cdot \mathbf{u} + \mathbf{M} \cdot \mathbf{v}) \\ &= -\mathbf{v} + \mathbf{F}(\mathbf{h} + \mathbf{M} \cdot \mathbf{v})\end{aligned}$$

Excitatory–inhibitory networks

Dale's law

... a neuron cannot excite some of its postsynaptic targets and inhibit others ...

$$\tau_E \frac{d\mathbf{v}_E}{dt} = -\mathbf{v}_E + \mathbf{F}_E(\mathbf{h}_E + \mathbf{M}_{EE} \cdot \mathbf{v}_E + \mathbf{M}_{EI} \cdot \mathbf{v}_I)$$

$$\tau_I \frac{d\mathbf{v}_I}{dt} = -\mathbf{v}_I + \mathbf{F}_I(\mathbf{h}_I + \mathbf{M}_{IE} \cdot \mathbf{v}_E + \mathbf{M}_{II} \cdot \mathbf{v}_I)$$

Linear system

– simplification making it solvable

$$\frac{d\mathbf{v}}{dt} = -\mathbf{v} + \mathbf{h} + \mathbf{M} \cdot \mathbf{v}$$

Consider the case when $\mathbf{M}$ is symmetric. The eigenvectors of $\mathbf{M}$ are orthogonal.

$$\mathbf{M} \cdot \mathbf{e}_\mu = \lambda_\mu \mathbf{e}_\mu \quad \mathbf{e}_\mu \mathbf{e}_\nu = \delta_{\mu\nu}$$

$$\mathbf{v}(t) = \sum_{\mu=1}^{N_\nu} c_\mu(t) \mathbf{e}_\mu, \quad \tau_r \frac{dc_\nu}{dt} = -(1 - \lambda_\nu) c_\nu(t) + \mathbf{e}_\nu \cdot \mathbf{h}$$

Solution

$$c_\nu(t) = \frac{\mathbf{e}_\nu \cdot \mathbf{h}}{1 - \lambda_\nu} \left(1 - \exp \frac{-t(1 - \lambda_\nu)}{\tau_r} + c_\nu(0) \exp \frac{-t(1 - \lambda_\nu)}{\tau_r} \right)$$

Steady state functions

$$\mathbf{v}_\infty = \sum_{\nu=1}^{N_\nu} \frac{\mathbf{e}_\nu \cdot \mathbf{h}}{1 - \lambda_\nu} \mathbf{e}_\nu$$

Selective amplification

When λ_1 is close to 1 while others are close to zero.

$$\mathbf{v}_\infty \approx \frac{\mathbf{e}_1 \cdot \mathbf{h}}{1 - \lambda_1} \mathbf{e}_1$$

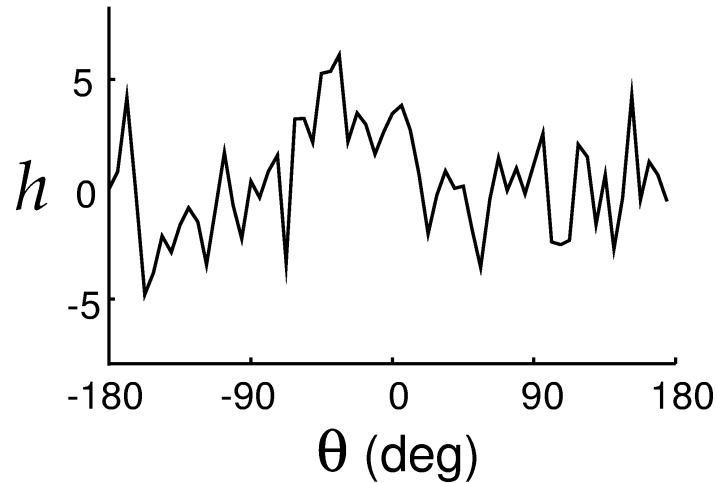
Input integration

When $\lambda_1 = 1$, others $\lambda_\nu \leq 1$.

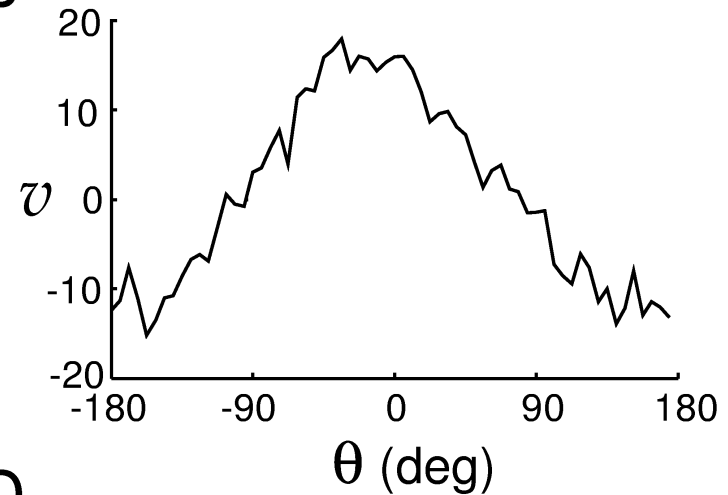
$$c_1(t) = c_1(0) + \frac{1}{\tau_r} \int_0^t dt' \mathbf{e}_1 \cdot \mathbf{h}(t')$$

Selective amplification example

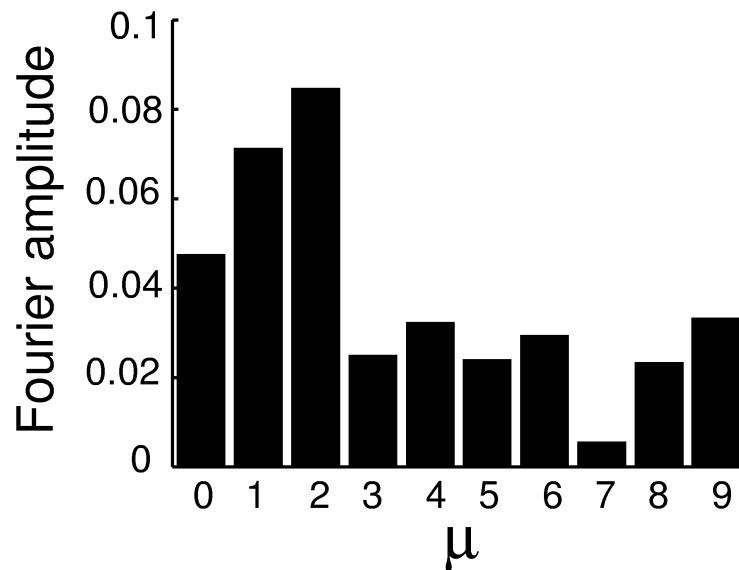
A



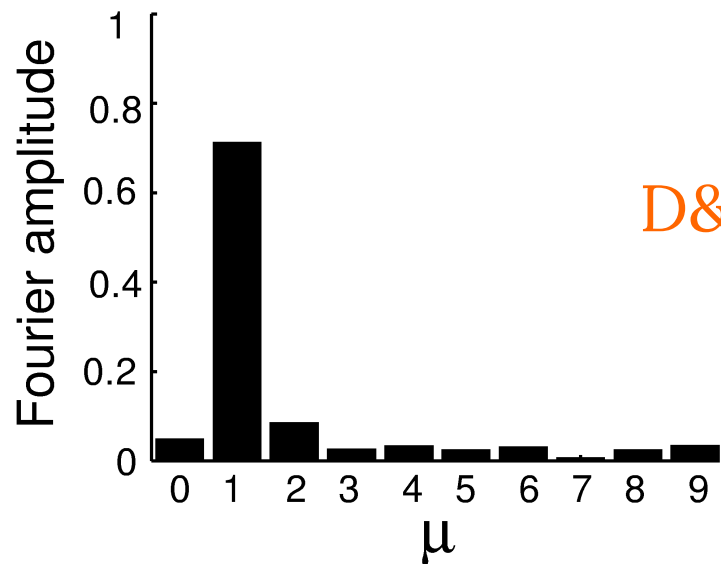
B



C



D

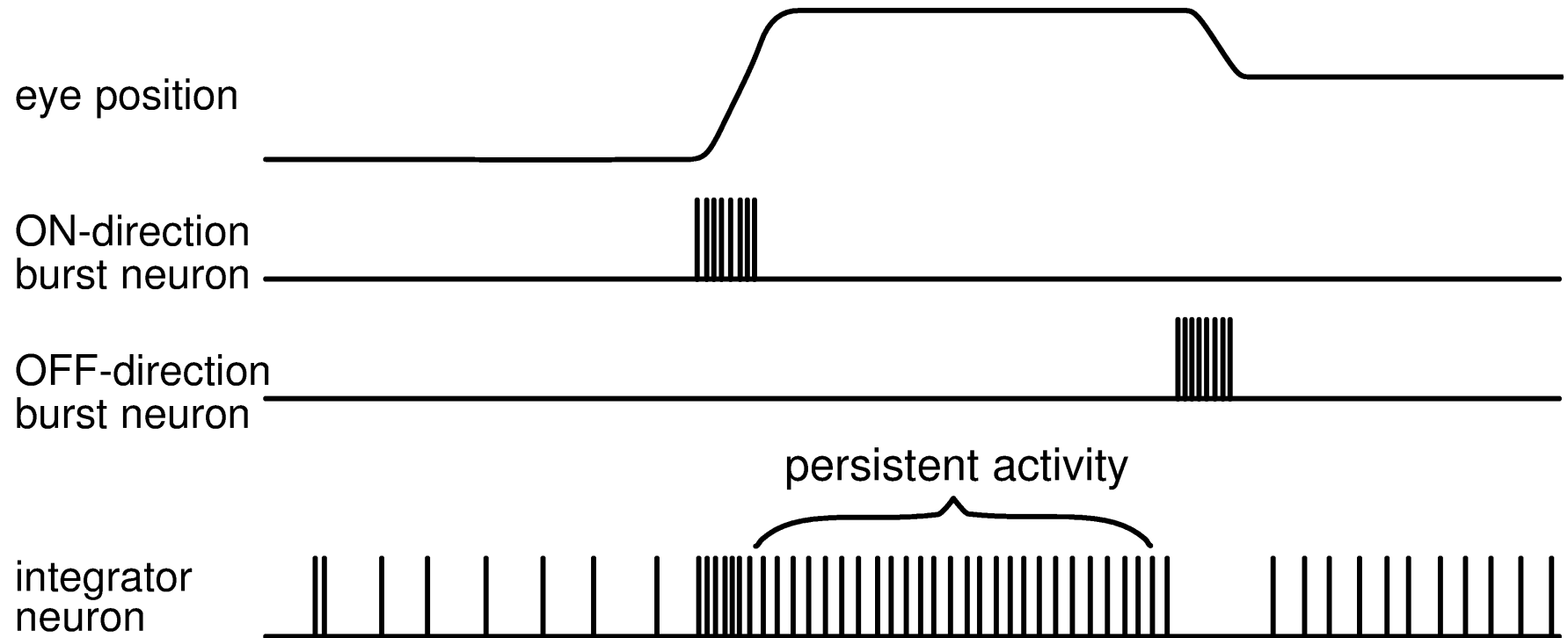


D&A Fig 7.8

Example use of an integrator

Horizontal eye positioning:

A form of
memory!



D&A Fig 7.7